

Emmanuel Alabi

Software Engineer & AI/Robotics Researcher

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Research Interests

Embodied AI; robot learning; computer vision; autonomous systems; multimodal intelligence. Specifically, how an agent's representation of a physical environment should be shaped by having to move through it, and when learned components genuinely outperform well-specified classical methods.

Broader interests: machine learning, robotics, state estimation, planning, reinforcement learning.

Education

Babcock University — BSc (Hons) Software Engineering 2019 — 2023

Second Class Upper Division, CGPA 4.00/5.00. Ilishan-Remo, Ogun State, Nigeria. School of Computing and Engineering Sciences.

Relevant coursework: Algorithms and Data Structures; Discrete Mathematics; Introductory Statistics; Introduction to Operations Research; Artificial Intelligence and Applications; Introduction to Big Data Engineering; Computer Organization and Assembly Language; Operating System I; Database System Design, Implementation and Management.

Research

Risk-Calibrated Semantic Navigation Under Distribution Shift 2026

Preregistered ROS 2 / Gazebo / Nav2 benchmark. doi.org/10.5281/zenodo.22181941

- Calibration improved and clean-route efficiency held, but the preregistered collision-reduction and severity hypotheses did not: better component calibration did not establish safer navigation.
- Four systems differing only in the treatment component; protocol, manifest, thresholds and analysis code hash-locked before execution. 6,840 episodes; independent clean-clone reproduction byte-identical.
- emmanuelalabi.com/research/risk-calibrated-semantic-navigation

Driver Drowsiness Detection — Reproduction & Evaluation Study 2026

Independent computer vision study. github.com/alabiemmanuel177/driver-drowsiness-reproduction

- An oracle-classifier decomposition showed the historical temporal threshold, not residual classification error, dominated false alerts once classifier accuracy was sufficient.
- Subject-disjoint reproduction on the MRL Eye Dataset; CNN trained from scratch against a HOG+SVM baseline, with deterministic splits, automated leakage checks and three-seed ablations.
- emmanuelalabi.com/projects/driver-drowsiness-reproduction

Research Training

Independent Research Programme — Embodied & Multimodal Autonomy 2026 — present

- Self-directed programme in the mathematical, machine-learning and robotics foundations for independent research: linear algebra, probability, optimisation, state estimation and rigid-body geometry, each implemented from first principles.
- Reproducing published baselines in perception and navigation before extending them.
- emmanuelalabi.com/research/research-programme-2026

Selected Academic Projects

BUCODEL — Learning Management System for Open & Distance e-Learning 2022 — 2023

Undergraduate Final-Year Project (SENG490, 72/100), Babcock University. Supervisor: Dr. Stephen Maitanmi.

- Co-designed and implemented a web-based learning management system for Babcock University's Centre for Open and Distance e-Learning, supporting student, lecturer and administrator workflows.
- Built course and resource management, assignments, authentication, user administration, collaborative course activity and role-specific interfaces using React, Node.js/Express and MongoDB.
- Conducted requirements analysis, system design, iterative implementation under a spiral methodology, and interface/usability, unit and integration testing.
- emmanuelalabi.com/projects/bucodel-learning-management-system

Selected Technical Experience

Founder / Lead Engineer — Obelo

- Production AI platform: multimodal generation, retrieval-augmented context, model routing across model families, asynchronous orchestration, and evaluation and quality systems.
- emmanuelalabi.com/projects/obelo-multimodal-ai-infrastructure

Software Engineer — SPay Business

- Production mobile-financial systems: authentication improvements, application performance, interface redesign, CI/CD automation and over-the-air mobile deployments.

Technical Skills

Research computing Python, PyTorch, NumPy, ROS 2, Gazebo, Nav2, statistical evaluation, uncertainty calibration, experiment design

AI systems Computer vision, multimodal systems, retrieval-augmented generation, LLM and model integration, evaluation systems

Software & infrastructure TypeScript, SQL, Docker, CI/CD, PostgreSQL, MongoDB, Node.js, React, Next.js

References

References available on request.